

Presentation Script

Read-aloud text for every slide of *DSDR_Final_Presentation.pptx*

Jhala Nath Kafle (081MSPSE009)

How to use this. Each block is one slide: the slide number, a small picture of the slide, and the paragraphs to read, numbered **Para 1, Para 2, ...** Read them in order, at a normal pace, and change the slide when the block ends. The brown lines ( *Point to*) are not read: they say where to point on the screen before the next paragraph.

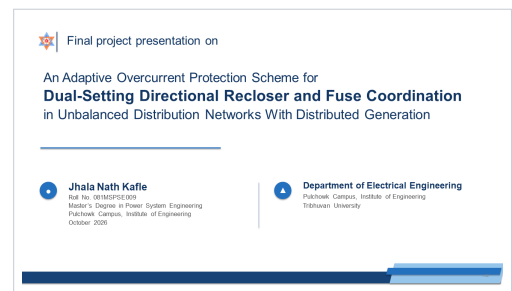
Timing. Slides 1 to 27 and the closing slide take about **9 minutes** to read at a calm speed (1132 words). The text is based on the speaker notes of the presentation, shortened a little to fit 8 to 10 minutes.

Numbers. Read “0.121” as “zero point one two one”, “4.16 kV” as “four point one six kilovolts”, “400E” as “four hundred E”, and “CTI” as the three letters.

SLIDE 1

Title

about 20 s to read



Para 1

Good afternoon. I am Jhala Nath Kafle, roll number 081 MSPSE 009. My project is an adaptive overcurrent protection scheme for dual-setting directional recloser and fuse coordination in distribution networks with distributed generation, on the IEEE 13-node feeder in PowerFactory.

SLIDE 2

Presentation Outline

about 5 s to read



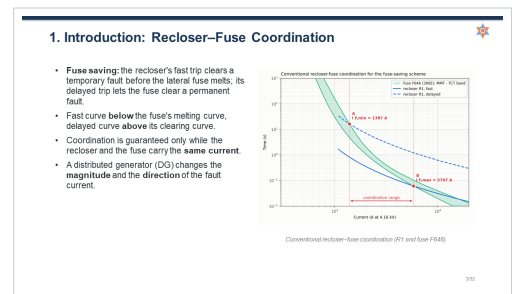
Para 1

I will cover the introduction, objectives, scope, methodology, results and conclusions.

SLIDE 3

1. Introduction: Recloser-Fuse Coordination

about 30 s to read



▷ Point to: the graph on the right: fast curve below the green fuse band, dashed delayed curve above it

Para 1

In a fuse-saving scheme the recloser trips first on its fast curve, so a temporary fault clears and the fuse is saved. For a permanent fault, the delayed curve lets the fuse clear its lateral.

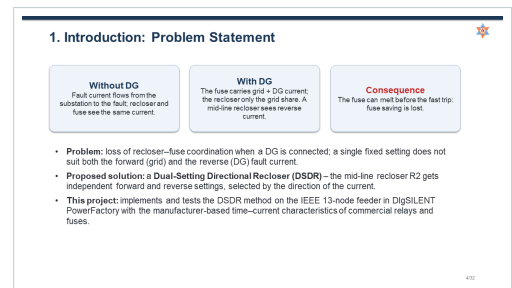
Para 2

This works only while the recloser and the fuse carry the same current, and a DG changes both the size and the direction of the fault current.

SLIDE 4

1. Introduction: Problem Statement

about 30 s to read



▷ Point to: the three boxes, left to right

Para 1

With a DG, the fuse carries grid plus DG current, while the recloser sees only the grid part, and a mid-line recloser can even see reverse current. The fuse may then melt before the fast trip.

Para 2

The solution studied is the dual-setting directional recloser, or DSDR: the mid-line recloser R2 gets separate forward and reverse settings, chosen by the direction of the current.

SLIDE 5

2. Objectives

about 30 s to read

2. Objectives

General Objective

To build and validate a DigSILENT PowerFactory implementation of the dual-setting directional recloser (DSDR) coordination method on the IEEE 13-node feeder, with and without distributed generation.

Specific Objectives

1. Build and verify the IEEE 13-node feeder with its recloser and fuse protection.
2. Perform unbalanced load-flow and short-circuit studies for LG, LL, LLG and LLL faults.
3. Calculate the recloser pickups, time dials and fuse coefficients.
4. Establish the coordination without DG, then with a 4.05 MVA synchronous DG at node 692.
5. Design the DSDR settings of R2, classify the resulting coordination, find the operating times.
6. Verify the reclosing sequence in the time domain (EMT simulation).
7. Study the effect of DG penetration from 0 to 100 %.

Para 1 The general objective was to build and validate a PowerFactory implementation of the DSDR method on the IEEE 13-node feeder, with and without DG.

Para 2 The seven specific objectives are: the model, load flow and short circuit, the settings, coordination without and with the DG, the dual setting of R2, a time-domain check, and the effect of DG penetration.

SLIDE 6

3. Scope and Limitations

about 30 s to read

3. Scope and Limitations

Scope

- IEEE 13-node feeder, 4.16 kV, unbalanced.
- LG, LL, LLO and LLL faults at 12 locations: 39 node/fault-type cells.
- Pickup, time-dial and fuse coordination calculations.
- DSDR forward and reverse settings; coordination with and without the DSDR.
- One EMT simulation of the reclosing sequence; DG penetration 0–100 %.

Limitations

- The method does not uniquely specify the R2 settings; fuse sizes or fault type of each operating-time case; they were defined in this study and documented.
- The CDG34 library relay is not directional; two relay units, direction from the fault location.
- Zero-margin criterion: $CTI = L_{MMT} - L_F > 0$; practical margins evaluated separately.
- One DG location (692); IEEE 34-node feeder not modelled.

Para 1 The study covers four fault types at twelve locations, which gives 39 node and fault-type cells, plus one EMT simulation and a penetration study.

▷ *Point to: the right column, Limitations*

Para 2 The limitations: some settings are not specified by the method and were defined here; the CDG34 relay model is not directional, so two relay units are used; a zero margin is used; and only one DG location is studied.

SLIDE 7

4. Methodology: Coordination Method

about 20 s to read

4. Methodology: Coordination Method

- A. Conventional scheme without DG: pickups by $I_p = OLF \times I_{nom}$ ($OLF = 1.25$), fuses sized and checked.
- B. Same settings with the DG connected: classification of all 39 cells.
- C. R2 as a DSDR: reverse pickup from the reverse load current; forward and reverse dials.
- If coordination is lost: revise the time dial; at the dial limit, upgrade the fuse size.
- Check: fuse melting time must exceed the recloser fast time; fuse clearing time below the delayed time.



▷ *Point to: the flowchart on the right*

Para 1 The method has three stages: A, the conventional design without DG; B, the same settings with the DG; and C, R2 as a DSDR. If coordination is lost, the time dial is revised first, and at the dial limit the fuse is made larger.

SLIDE 11

4. Methodology: Parameters of the Synchronous DG

about 5 s to read

4. Methodology: Parameters of the Synchronous DG					
Parameter	Symbol	Value	Parameter	Symbol	Value
Leakage reactance	X_l	0.01 pu	d-axis subtransient reactance	X'_{ld}	0.118 pu
Stator resistance	R_a	0.0014 pu	q-axis transient OC time const.	T'_{dq}	1.25 s
d-axis synchronous reactance	X_{sd}	1.4 pu	q-axis subtransient OC time const.	T''_{dq}	0.19 s
d-axis transient reactance	X'_{sd}	0.231 pu	Mechanical damping time	$M = 2H$	1.5 s
d-axis subtransient reactance	X''_{sd}	0.118 pu	DG rating	S_n	4.05 MVA
d-axis transient OC time const.	T'_{sd}	5.5 s	Stop-up transformer	U_h	0.69 / 4.16 kV
d-axis subtransient OC time const.	T''_{sd}	0.05 s	Transformer leakage reactance	x_T	0.15 pu
q-axis synchronous reactance	X_{sq}	1.372 pu	Connection node		692
q-axis transient reactance	X'_{sq}	0.8 pu			

Dynamic synchronous machine in PowerFactory, operating point 2.24 RME, voltage control at 1.0 pu

Para 1 These are the parameters of the synchronous DG, 4.05 MVA, connected at node 692.

SLIDE 12

4. Methodology: Protection Settings

about 20 s to read

4. Methodology: Protection Settings			
Device	Characteristic	Pickup / plug (fast / delayed)	Dial (fast / delayed)
R1 (feeder head)	GE IAC extremely inverse	720 A	TDS 0.5 / 10
R2 forward (grid current)	CDG34	300 / 600 A	TMS 0.1 / 1.0
R2 reverse (DG current)	CDG34, CT 500/5	150 / 300 A	TMS 0.1 / 1.0

- Reverse load current of R2 with the DG: 252 A → reverse pickup 315 A (OLF = 1.25).
- Dial revision could not help: R1 and R2 are already at the ends of their dial ranges.
- Fuse revision (step 9 of the method): F632 → 500E; F633, F646, F-DL, F671-1 → 400E; F662-R → 250E.

► Point to: the table

Para 1 R1 has a pickup of 720 amperes. R2 forward uses plugs of 300 and 600 amperes, and the new reverse group 150 and 300 amperes.

Para 2 The dials were already at their limits, so the fuse sizes were revised as shown.

SLIDE 13

5. Results: Branch Currents and Fault Levels (DG out)

about 15 s to read

5. Results: Branch Currents and Fault Levels (DG out)			
Branch	Rated current I _{nom} (A)	I _{min} (kA) L5 3 G, farthest node	I _{max} (kA) bolted, nearest node
6066-6102	587.7	1.07	4.73
6102-6103	81.0	0.76	4.1
6102-6460	143.1	0.75	3.42
6102-6171	470.0	0.86	3.36
307M 1 HV side	81.0	0.88	1.81
307M 1 LV side	766.4	0.71	15.67
645-646	66.1	0.73	3.11
6171-6802	239.7	0.74	3.39
6171-6804	71.1	0.66	2.66
6171-680	0.0	0.64	2.63
6802-675	205.9	0.74	3.15
684-611	71.1	0.68	1.96
684-652	63.0	0.66	1.87

Unbalanced load flow (targeted phase) and short circuit, complex method, 307M 1 LV side at 0.45 kV

Para 1 The first result is the base case without DG: the rated current and the minimum and maximum fault current of each branch. The feeder head sees 4.73 kiloamperes.

SLIDE 14

5. Results: Fuse Coefficients

about 10 s to read

5. Results: Fuse Coefficients (eq. 9, $a = -1.8$)

Fuse	IP (kA)	I/a	t fast (s)	t delayed (s)	I fuse (kA)	h t	Installed fuse	t MDT (s)
F002	3418	20	0.133	2.070	1.824	0.02	400E	0.012
F003	4032	15	0.100	2.000	1.954	0.03	250E	0.100
F004	15671	12	0.439	8.772	3.277	0.08	40-400A	0.021
F045	3418	12	0.133	2.070	0.979	0.35	250E	0.158
F046	3712	12	0.166	2.116	1.142	0.35	300E	0.273
F-CL	3945	15	0.100	2.150	1.118	0.52	250E	0.112
F071	2360	15	0.054	1.601	0.827	0.27	250E	0.161
F071-1	2360	30	0.054	1.601	1.214	0.44	250E	0.278
F082	3390	12	0.054	1.601	0.909	0.11	200E	0.084
F082-R	3551	20	0.057	1.609	0.943	0.27	200E	0.112
F075	3511	15	0.057	1.609	0.930	0.5	200E	0.112
F075-2	3551	30	0.069	2.572	1.946	0.45	300E	0.458
F084	1857	20	0.113	5.778	2.946	0.35	200E	0.465
F011	1857	15	0.113	5.778	1.530	0.07	125E	0.120
F002	1878	12	0.112	5.660	1.981	0.18	100E	0.207

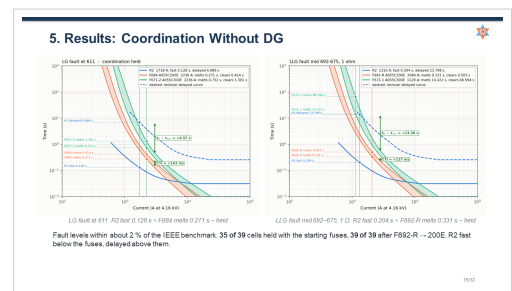
h: largest fault current before the fuse, CG out, I/a: position in a series of fuses (1 = closest to the fault)

Para 1 These are the fuse coefficients from the method's fuse equation, with the installed fuse and its melting time in the last columns.

SLIDE 15

5. Results: Coordination Without DG

about 15 s to read



Para 1 Without DG, 35 of 39 cells were coordinated, and all 39 after one fuse change.

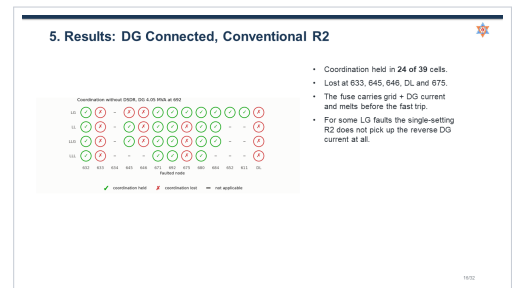
▷ Point to: the left plot, then the right plot

Para 2 In both examples, R2's fast curve is below the fuses and its delayed curve is above them.

SLIDE 16

5. Results: DG Connected, Conventional R2

about 20 s to read



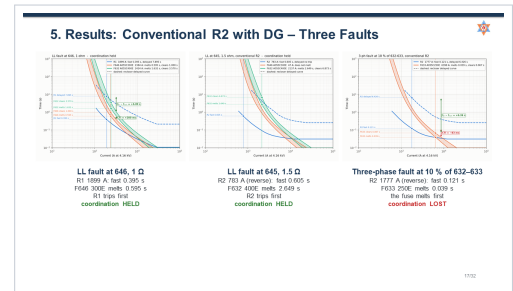
▷ Point to: the red crosses in the chart

Para 1 With the DG and unchanged settings, coordination holds in only 24 of 39 cells. The fuse melts before the fast trip, and for some line-to-ground faults R2 does not even pick up the reverse current.

SLIDE 17

5. Results: Conventional R2 with DG - Three Faults

about 20 s to read



▷ Point to: the right plot

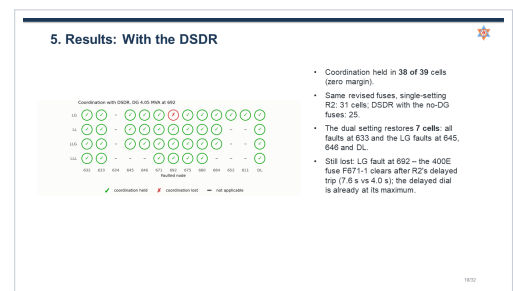
Para 1

Three faults with the conventional settings. In the first two the recloser still trips first. For the three-phase fault close to 632, R2 trips in 0.121 seconds, but the fuse melts in 0.039 seconds: coordination is lost.

SLIDE 18

5. Results: With the DSDR

about 35 s to read



▷ Point to: the chart: only one red cross is left

Para 1

With R2 as a DSDR and revised fuses, 38 of 39 cells are held under the zero-margin criterion. With the same revised fuses but a single-setting R2, only 31, so the dual setting itself restores seven cells.

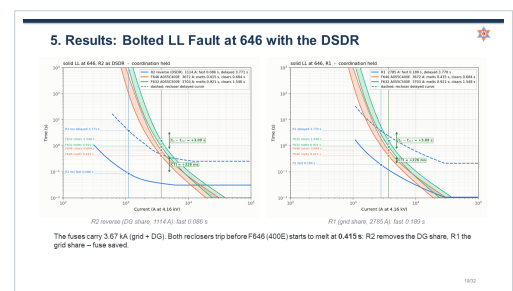
Para 2

The DSDR with the old fuses gives only 25, so it needs the fuse revision. Only the line-to-ground fault at 692 stays lost, limited by the maximum delayed dial of R2.

SLIDE 19

5. Results: Bolted LL Fault at 646 with the DSDR

about 20 s to read



▷ Point to: the left plot for R2, the right plot for R1

Para 1

A bolted line-to-line fault at 646 with the DSDR. R2 trips in 0.086 seconds on its reverse group and R1 in 0.189 seconds. Both are before the fuse starts to melt at 0.415 seconds, so the fuse is saved.

SLIDE 20

5. Results: Operating Times with the DSDR

about 10 s to read

5. Results: Operating Times with the DSDR (DG in)									
Node	Fault	I R1 (kA)	R1 test / delayed (s)	I R2 (kA)	R2 test / delayed (s)	Fuse	I fuse (kA)	Fuse MNT / TCT (s)	
632	L-L	4734	0.081 / 1.627	1813 rev	0.051 / 1.416	—	—	—	
633	L-L	3870	0.106 / 2.111	1500 rev	0.060 / 2.001	F033	5420	0.149 / 0.235	
640	L-L	3524	0.100 / 2.006	1200 rev	0.071 / 2.002	F032	4491	0.613 / 0.819	
646	L-L	3785	0.189 / 3.770	1114 rev	0.080 / 3.771	F046	3672	0.415 / 0.604	
DL	L-L	3864	0.100 / 2.125	1072 rev	0.048 / 1.215	F-L	5802	0.121 / 0.193	
671	L-L	3412	0.134 / 2.677	3381 led	0.054 / 1.000	—	—	—	
682	L-L	3412	0.134 / 2.677	3381 led	0.054 / 1.000	F071.5	3381	0.525 / 0.881	
675	L-L	3680	0.104 / 2.082	3660 led	0.059 / 1.044	F062.0	6032	0.060 / 0.106	
684	L-L	3543	0.222 / 4.437	3543 led	0.072 / 2.813	F074.2	3603	0.149 / 0.230	
680	L-L	2789	0.188 / 3.761	2757 led	0.066 / 2.381	—	—	—	
611	L-G	1814	0.432 / 8.709	1718 led	0.126 / 0.386	F084	2236	0.271 / 0.414	
652	L-G	1795	0.448 / 8.919	1781 led	0.125 / 0.570	F052	2259	0.136 / 0.195	

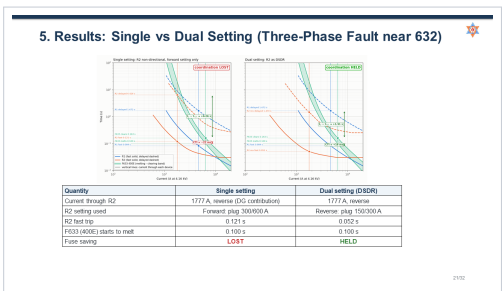
Busfed L-L, L-L at two-phase, L-G at one-phase nodes. Every fuse starts to melt after the last fast trip; smallest margins 6 ms (672) and 15 ms (674)

Para 1 This table gives the operating times with the DSDR. In every row the fuse starts to melt after the last fast trip.

SLIDE 21

5. Results: Single vs Dual Setting

about 20 s to read



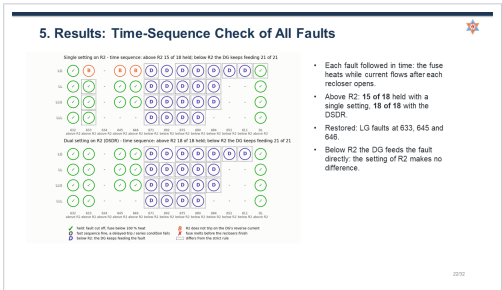
▷ Point to: the table under the plots

Para 1 One case in detail: the three-phase fault near 632. With the single setting, R2 trips in 0.121 seconds, after the fuse starts melting: lost. With the reverse group, R2 trips in 0.052 seconds: held. Only the setting group changed.

SLIDE 22

5. Results: Time-Sequence Check of All Faults

about 20 s to read

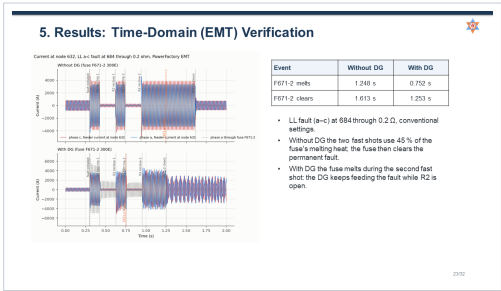


Para 1 As a second check I followed every fault in time. Above R2 the single setting holds 15 of 18 cells and the DSDR all 18. Below R2 the DG feeds the fault directly, so R2's setting makes no difference.

SLIDE 23

5. Results: Time-Domain (EMT) Verification

about 25 s to read



▷ Point to: the upper waveform = without DG, the lower = with DG

Para 1

This EMT simulation is a line-to-line fault at 684. Without DG, the two fast shots use only 45 per cent of the fuse’s melting heat, and the fuse then clears the permanent fault.

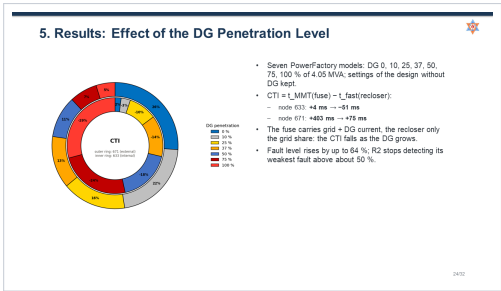
Para 2

With the DG, current keeps flowing while R2 is open, and the fuse melts at 0.75 seconds, during the second fast shot.

SLIDE 24

5. Results: Effect of the DG Penetration Level

about 15 s to read



▷ Point to: the ring chart

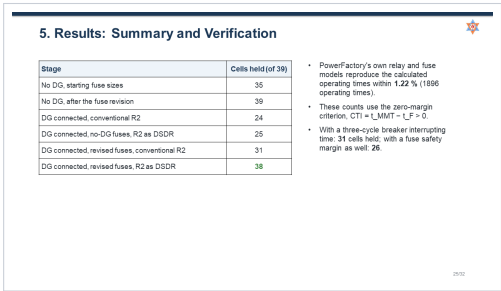
Para 1

Seven models with the DG from 0 to 100 per cent show the CTI falling: at 633 from plus 4 to minus 51 milliseconds, at 671 from 403 to 75 milliseconds.

SLIDE 25

5. Results: Summary and Verification

about 30 s to read



▷ Point to: the table, top to bottom

Para 1

In summary: without DG, 35 cells and 39 after the fuse revision; 24 with the DG and a conventional R2; 25 with the DSDR alone, 31 with the fuse revision alone, and 38 with both.

Para 2

These counts use a zero margin; with a three-cycle breaker time they become 31, and 26 with a fuse safety margin as well.

SLIDE 26

6. Conclusions and Future Work

about 35 s to read

Para 1

To conclude: with the DG, a conventional R2 keeps only 24 of 39 cells. The combined DSDR and fuse revision raises this to 38 of 39, and the last cell is limited by R2's maximum delayed dial.

Para 2

The DSDR alone gives only 25, so it is necessary but not sufficient, and faults below R2 remain a limit. Future work includes the 34-node feeder, a true directional element, practical margins and inverter-based DG.

6. Conclusions and Future Work

- The DSDR method was implemented on the IEEE 13-node feeder in DigSILENT PowerFactory with manufacturer-based relay and fuse characteristics; the model meets the IEEE benchmark within about 2 %.
- Without DG, 35 of 39 cells with the starting fuses, 39 of 39 after the fuse revision.
- With the DG and a conventional R2, DSDR alone: 25; fuse revision alone: 31; combined DSDR and fuse revision: 38 of 39 (zero margin). The remaining LG fault at 692 is limited by R2's maximum delayed dial.
- With a three-cycle breaker time the 38 cells become 31, and 28 with a fuse safety margin as well.
- The DSDR is necessary but not sufficient; it needs the fuse revision, and faults below R2 remain a limit. PowerFactory's relay and fuse models confirm the times within 1.22 %.

Future work

- IEEE 34-node feeder; explicit directional element for R2; practical margins; setting-group change triggered by the DG state; inverter-based DG; hardware-in-the-loop tests.

20/22

SLIDE 27

7. References

about 0 s to read

7. References

- [1] M. Yousef, A. Jallian, K. M. Mulraj, D. Sutanto, "An adaptive overcurrent protection scheme for dual-setting directional recloser and fuse coordination in unbalanced distribution networks with distributed generation," IEEE Trans. Ind. Appl., vol. 50, no. 2, pp. 1831-1842, 2020.
- [2] W. H. Karsting, "Radial distribution test feeders," IEEE Trans. Power Syst., vol. 6, no. 3, pp. 975-985, 1991.
- [3] G. Bermudez et al., "IEEE standard inverse-time characteristics equations for overcurrent relays," IEEE Trans. Power Del., vol. 14, no. 3, pp. 868-872, 1999.
- [4] A. Yousef et al., "A classification technique for recloser-fuse coordination in distribution systems with distributed generation," IEEE Trans. Power Del., vol. 27, no. 1, pp. 176-185, 2012.
- [5] M. Yousef, K. M. Mulraj, D. Sutanto, "Overcurrent protection scheme for the IEEE 13-node benchmark test feeder with improved selectivity," IEEE PES General Meeting, 2020.
- [6] W. H. Karsting, G. Shiva, "Short circuit analysis of IEEE test feeders," IEEE PES T&D, 2012.
- [7] DigSILENT GmbH, PowerFactory 2021 User Manual, 2021.
- [8] GE Multilin, 750737 Feeder Protection Relay - Instruction Manual (GEK-106291F), 2010.

21/22

Para 1

These are the main references.

SLIDE 32

Thank You

about 5 s to read

Para 1

Thank you for your attention. I am happy to take your questions.

Thank You

Questions and Discussion

22/22

Appendix slides — use only if you are asked

Slides 28 to 31 come after the references. Do not present them; open one when a question needs it.

SLIDE 28

Appendix: Comparison with the Reference Paper

about 30 s to read

Para 1

Compared with the reference paper, load currents agree within 2 per cent and the R1 worked example almost exactly. My fault levels are lower because I validated against the IEEE benchmark. The paper reports 30 and 39 coordinated cells, I obtained 24 and 38. The main finding is reproduced: the DSDR with the fuse revision restores the coordination between the grid and the DG.

Appendix: Comparison with the Reference Paper			
Aspect	This project	Reference paper [1]	Assessment
Rated branch currents	within 2 %	Table II	Very good
Fault levels	IEEE benchmark, 4.73 kA at head	5.41 kA, some branches higher	Lower
R1 worked example (4219 A)	0.006 / 1.502 s	0.007 / 1.502 s	Very good
Coordination with DG, conventional R2	24 of 39 cells	30 of 39 (Fig. 14)	Good (29 cells agree)
Coordination with the DSDR	38 of 39 cells	39 of 39 (Fig. 17)	Good: one cell at dial limit
Fuse current, LL at 646	3.67 kA	3.64 kA (Fig. 15)	Very good
R2 reverse current, LL at 646	1.11 kA	1.54 kA	Lower
EMT fuse operation	melts 1.25 s, clears 1.61 s	1.21 s (Fig. 16)	Close
DG model	Dynamic synchronous machine	Synchronous machine	Same

SLIDE 29

Appendix: Fault Levels - IEEE Benchmark, This Study and the Paper

about 30 s to read

Para 1

My fault levels are lower than the paper's, so I checked them against the IEEE short-circuit benchmark. My model agrees within 2 per cent at ten of eleven nodes. The paper's values are up to 151 per cent higher, and two of them exceed its own feeder-head value of 5.41 kiloamperes, which is not possible in a radial feeder without DG.

Appendix: Fault Levels – IEEE Benchmark, This Study and the Paper					
Node	IEEE benchmark (kA)	This study (kA)	Study vs benchmark	Paper, Table I (kA)	Paper vs benchmark
632	4.80	4.74	-1.3 %	5.41	+12.7 %
633	4.15	4.10	-1.1 %	6.73	+62.2 %
646	5.41	3.42	-36.2 %	4.25	-21.6 %
646	3.86	3.71	-3.9 %	3.64	-4.7 %
671	3.35	3.36	+0.3 %	4.52	+34.9 %
682	3.35	3.38	+1.2 %	3.09	-8.1 %
675	3.12	3.15	+1.0 %	7.83	+151.0 %
680	2.91	2.80	-3.8 %	6.25	+114.8 %
684	2.82	2.86	+1.5 %	3.47	+23.4 %
682	1.79	1.87	+4.5 %	2.67	+49.2 %
611	1.85	1.86	+0.4 %	1.93	+4.3 %

Maximum fault current, DG out. Benchmark: Kundur & Stovall [8], values as quoted in [2]. Paper: K. Iwasaki et al. (2004) IEEE Trans. on Power Delivery, 19(2), 1011–1018.

SLIDE 30

Appendix: DSDR Operating Times - This Study and the Paper

about 25 s to read

Para 1

The operating times with the DSDR show the same behaviour as the paper: in every row the fuse melts after the fast trip of the recloser that protects it. R1's delayed time is twenty times the fast time in both, which confirms the dials. R2's settings are not published, so its times are compared in trend only.

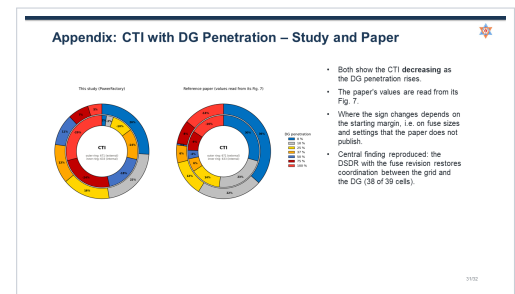
Appendix: DSDR Operating Times – This Study and the Paper					
Node	Fault	R1 fast (s)	R1 delayed (s)	R2 fast (s)	R2 delayed (s)
632	LLL	0.081 (0.07)	1.627 (1.386)	0.051 (0.045)	1.416 (0.514)
633	LLL	0.106 (0.084)	2.111 (1.88)	0.060 (0.052)	2.001 (0.705)
646	LL	0.163 (0.140)	3.080 (2.886)	0.075 (0.067)	3.002 (1.698)
646	LL	0.169 (0.138)	3.779 (3.555)	0.080 (0.103)	3.771 (2.946)
671	LLL	0.106 (0.087)	2.123 (1.733)	0.040 (0.076)	1.213 (1.402)
671	LLL	0.134 (0.105)	2.827 (2.081)	0.054 (0.088)	1.600 (1.113)
680	LLL	0.124 (0.110)	2.877 (2.207)	0.054 (0.072)	1.600 (1.238)
675	LL	0.150 (0.127)	3.152 (2.414)	0.069 (0.074)	1.941 (1.332)
684	LL	0.222 (0.125)	4.437 (2.507)	0.072 (0.075)	2.813 (1.384)
680	LLL	0.168 (0.137)	3.161 (2.747)	0.060 (0.079)	2.081 (1.555)
611	LL	0.470 (0.336)	8.788 (6.923)	0.138 (0.140)	6.965 (4.385)
682	LL	0.440 (0.376)	8.919 (7.525)	0.123 (0.143)	6.976 (4.37)

Each row: this study / reference paper [1]. DG connected, R2 on (DSDR).

SLIDE 31

Appendix: CTI with DG Penetration - Study and Paper

about 10 s to read



Para 1 Both studies show the margin decreasing with DG penetration; where it turns negative depends on settings the paper does not publish.